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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Subject : COMP4422 Computer Graphics

Session : 2024 / 2025 Semester I

Date : 7 December 2024 Time : 12:30 - 14:30

Time Allowed: 2 Hours

Subject Examiner(s): Dr TANG Kai Tai Jeff

This question paper has a total of ___6___ pages (cover included).
(Some pages may be intentionally omitted.)

Instructions to Candidates:

- The exam includes TEN multiple-choice items and SIX open-ended questions.
- Attempt ALL questions.
- Write reasonable, neat, and coherent answers in the provided answer book.
- Write down your name and student number on the cover page of the answer book and on the top of each extra paper (if any).
- Request extra or draft papers if needed.
- No books, articles, printouts, or cheat sheets are allowed.
- No electronic equipment except HKEA-approved calculators without pre-stored programs.
- Don't take away the question paper. The writing on the question paper will not be graded.

Do not turn over the page until you are told to do so!

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Section B Short Questions (40 marks)

Answer **ALL FOUR** questions in this section. Write down your answers in the **answer book** provided.

Question B1. [10 marks]

Describe the responsibility of the vertex shader, rasterizer and pixel shader stage of the graphics pipeline.

Question B2. [10 marks]

Polygon meshes are the most common representation of geometric shapes in computer graphics, and many algorithms rely on them. Describe a method for calculating the normal vector at each vertex of a polygon mesh, made up of polygons with varying number of sides. Include the mathematical approach for computing the normal based on the positions of the vertices in the adjacent polygons. While explaining, draw the vertices to visualize how you arrived at the answer.

Question B3. [10 marks]

Consider in the following given surface, the surface normal N, the direction of the point light source L, the direction of the reflected light R, and the direction to the eye V.

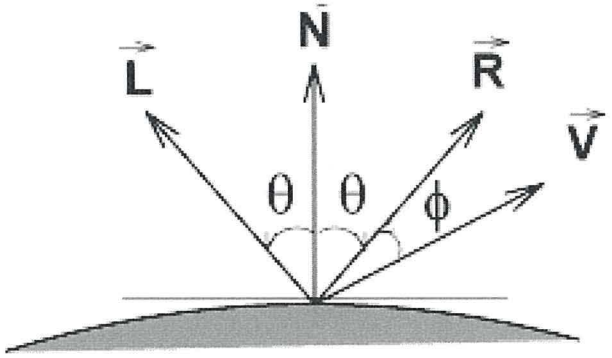


Figure B3. A surface of an object to be rendered.

(a) Find the equation for diffuse reflection. Include more parameters that may be beneficial. Explain succinctly the mechanics of diffuse reflection. Finally, provide an example of a substance whose appearance may be modelled with diffuse reflection. (6 Points).

(b) As a follow-up question of Part (a), what light component will be adjusted, what illumination model will be used, and how will the parameters be set so that the surface discussed in Part (a) becomes a perfect mirror? (4 marks)

Question B4. [10 marks]

What is the main problem with utilizing Béizer curves to represent 3D models? (2 marks). How might the B-Spline approach help with this problem? (4 marks) Furthermore, discuss two geometric characteristics of 3D objects that lead people to store them as meshes rather than curves. (4 marks)

End of Section B

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Section C Long Questions (40 marks)

Answer **ALL TWO** questions in this section. Write down your answers in the **answer book** provided.

Question C1. [20 marks]

Assume you are a player walking through a detailed 3D game world populated by buildings, trees, and other characters. As you explore this environment, think about how a non-player character (NPC) would perceive the surroundings.

(a) Describe the process by which the environment's 3D coordinates are transformed into the 2D image displayed on the monitor in five steps. In your explanation, explain how the NPC's viewpoint is determined, how depth and perspective are represented on the screen, and how only relevant parts of the scene are rendered for display. Use your knowledge of the graphical rendering process to inform your response. (10 marks)
Hint: consider the following keywords:
Clipping and Viewport Transformation, Projection Transformation, Modeling Transformation, Normalization Transformation, Viewing Transformation

(b) What is a z-buffer and how does it work in computer graphics? (10 marks)

Question C1. [20 marks]

(a) Even though the world is three-dimensional, 4x4 matrices are used in computer graphics to represent transformations. Why are 4x4 matrices used instead of 3x3 matrices? (5 marks)

(b) Given the three matrices of different operations **J**: translation along the vector $v = (2, 0, 4)$, **K**: +90-degree rotation about the z-axis and **L**: non-uniform scaling with 2 in x, 4 in y and 8 in z.

i) Give the (4×4) matrix form of **J**, **K** and **L**. (3 marks)

ii) Given the point $P = (1, 0, 2, 1)$, calculate the location of $P_0 = \mathbf{LJK}P$? (6 marks)

iii) Describe the transformation as (a set of) matrix operations required to transform the triangle from T_1 to T_2, as illustrated in the figure below. (6 marks)

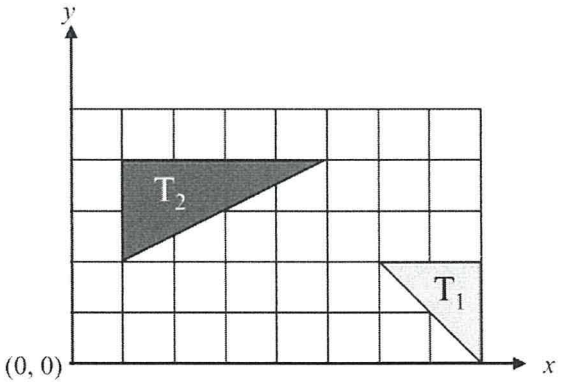


Figure C1. Transform from T_1 to T_2

End of Examination